# Armando Castillo

**Portfolio Project**

*Campaing Analysis*

LinkedIn GitHub

# Introduction

## About me

- Data Analyst focused on turning raw data into actionable insights.
- Experience in SQL, dashboards, reporting, and business analysis.
- I help clients understand performance, identify trends, and make data-driven decisions.
- Skilled in tools such as SQL, Excel, Jupyter Notebooks, Databricks, Tableau / Power BI / Looker Studio.
- Worked in many fields of study: Government, Banking, Marketing, HR and more.

# Social Media Campaign Analysis Overview

## Data preparation

Reviewed the three datasets, checked inconsistencies, and standardized key fields.

## Data Analysis

Built a SQL query to join all three sources into one unified dataset for analysis. Calculated KPIs for analysis.

## Data Visualization

Created the dashboard to compare channels, track KPIs, and identify performance opportunities.

# Data Preparation

First, I focused on **data preparation**, where I reviewed the three source datasets individually to identify inconsistencies, missing values, and field differences across platforms.

After validating and standardizing the structure, I created a **SQL query** to join the three datasets into a **unified table for analysis and exploration**.

This query is included in the **GitHub repository** as part of the project documentation. Once the data was consolidated, I moved into the final phase: exploratory analysis and dashboard development, where I used the unified dataset to compare performance across channels and build KPI-driven insights.

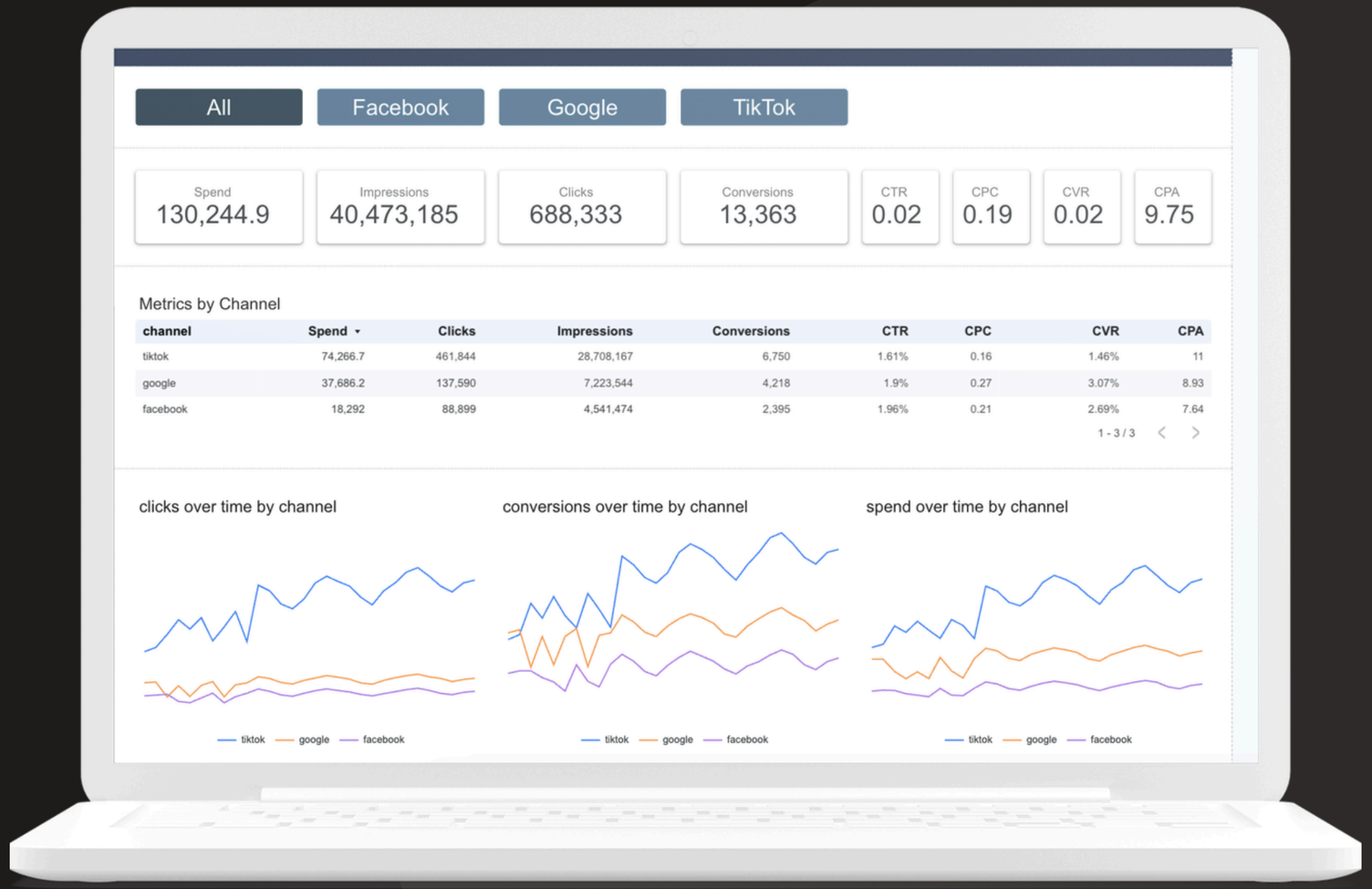
# Data Analysis

Across all paid media channels, the campaign mix generated 130.2k in spend, 40.5M impressions, 688.3k clicks, and 13.4k conversions.

Overall efficiency was solid, with a blended:

- **CTR of 1.7%**
- **CPC of 0.19**
- **CVR of 1.94%**
- **CPA of 9.75.**

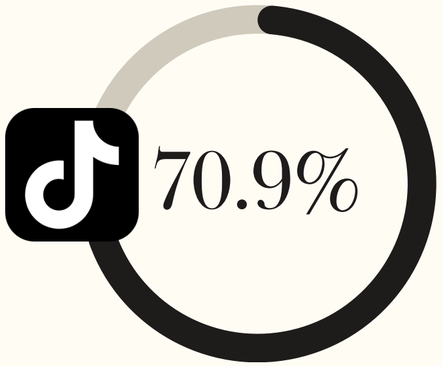
The data shows a clear split in channel roles.




| channel | Spend | Clicks | Impressions | Conversions | CTR | CPC | CVR | CPA |
|---|---|---|---|---|---|---|---|---|
| tiktok | 74,266.7 | 461,844 | 28,708,167 | 6,750 | 1.61% | 0.16 | 1.46% | 11 |
| google | 37,686.2 | 137,590 | 7,223,544 | 4,218 | 1.9% | 0.27 | 3.07% | 8.93 |
| facebook | 18,292 | 88,899 | 4,541,474 | 2,395 | 1.96% | 0.21 | 2.69% | 7.64 |

# Overall Overview

TikTok is the main volume driver, contributing the largest share of spend, impressions, and clicks.




Impression % from all 3 platforms

Google is the strongest performance / efficiency channel, generating the highest conversion efficiency and strongest return.

Highest CVR
(Conversion Rate)

Facebook is the most cost-efficient converter, with the lowest CPA among the three channels.

Lowest CPA
(Cost Per Adquisition)

# Data Visualization

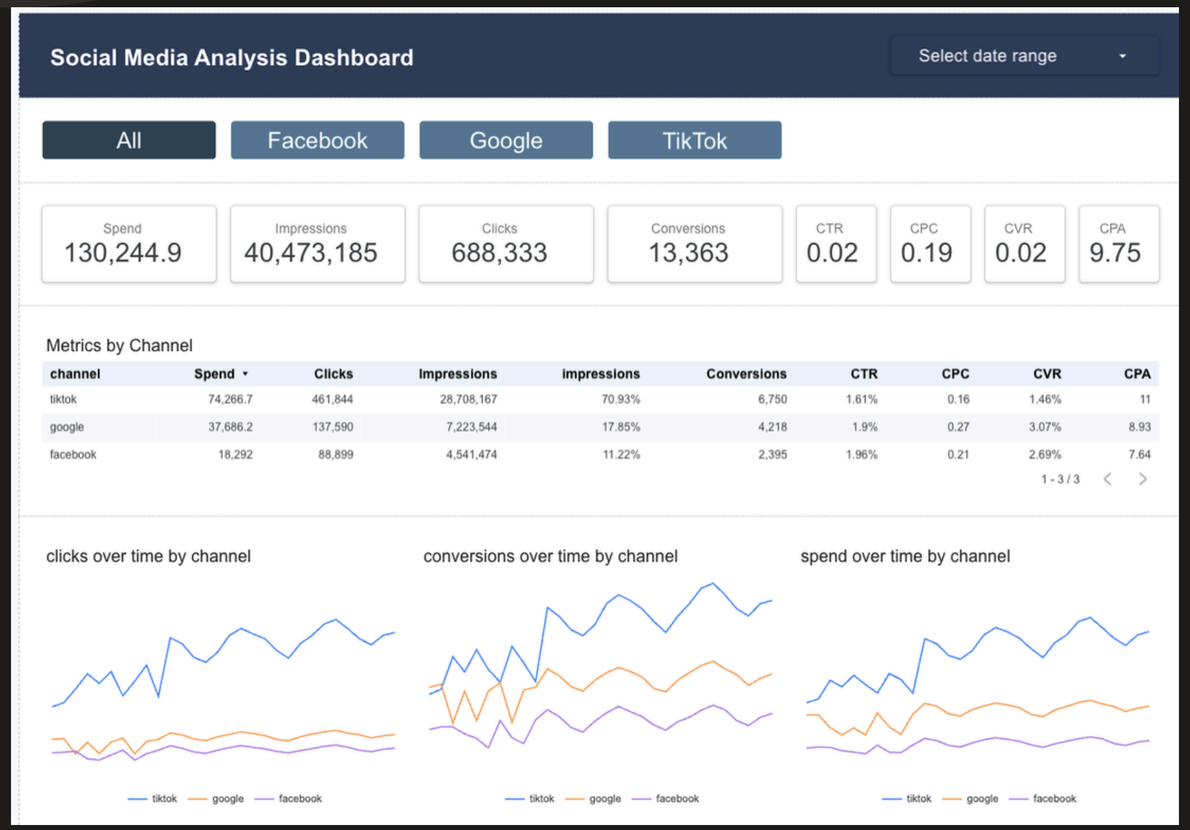
After consolidating the three datasets into a unified table, I designed the dashboard to make cross-platform performance easy to interpret.

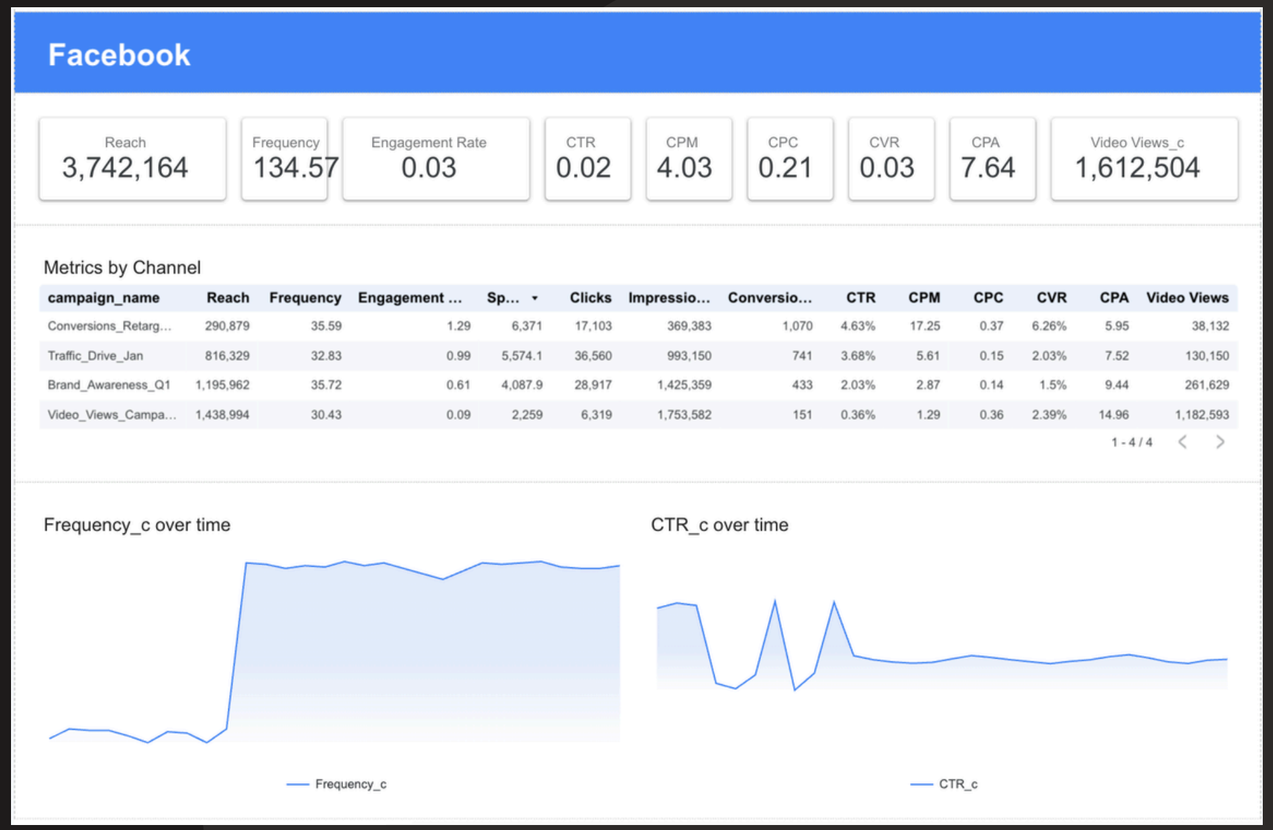
I selected key KPIs for a high-level summary and used scorecards, tables, and trend charts to compare channels, monitor performance over time, and highlight campaign-level results.

This approach turned the raw data into a clear, decision-focused view of overall performance and optimization opportunities.

You can see it in the **Github repository**.

# Data Visualization



# Data Visualization

## Google

| Spend | Conversions | Conversion Value | ROAS | CTR | CPC | CVR | CPA |
|---|---|---|---|---|---|---|---|
| 37,686.2 | 4,218 | 210,900 | 5.6 | 0.02 | 0.27 | 0.03 | 8.93 |

| Search Impression Share | Quality Score |
|---|---|
| 0.6 | 7.43 |

### Metrics by Channel

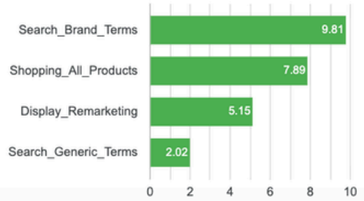
| campaign_name | Spend | Search Impression Sh... | Quality Score | ROAS | Clicks | Impressions | Conversions | CTR | CPC | CVR | CPA |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Search_Generic_Terms | 15,548.9 | 12 | 6.79 | 2.02 | 24,162 | 1,215,240 | 627 | 1.99% | 0.64 | 2.59% | 24.8 |
| Shopping_All_Products | 11,416.7 | 18.06 | 8 | 7.89 | 62,446 | 1,871,485 | 1,801 | 3.34% | 0.18 | 2.88% | 6.34 |
| Search_Brand_Terms | 7,368.1 | 27.42 | 9 | 9.81 | 38,603 | 740,093 | 1,445 | 5.22% | 0.19 | 3.74% | 5.1 |

1 - 4 / 4

### Participation by Campaign

| campaign_name | Spe... | spend | Conversions | conversions | Conversion Value | Conversion Value |
|---|---|---|---|---|---|---|
| Search_Generic_Terms | 15,548.9 | 41.26% | 627 | 14.86% | 31,350 | 14.86% |
| Shopping_All_Products | 11,416.7 | 30.29% | 1,801 | 42.7% | 90,050 | 42.7% |
| Search_Brand_Terms | 7,368.1 | 19.55% | 1,445 | 34.26% | 72,250 | 34.26% |
| Display_Remarketing | 3,352.5 | 8.9% | 345 | 8.18% | 17,250 | 8.18% |

### ROAS by Campaign

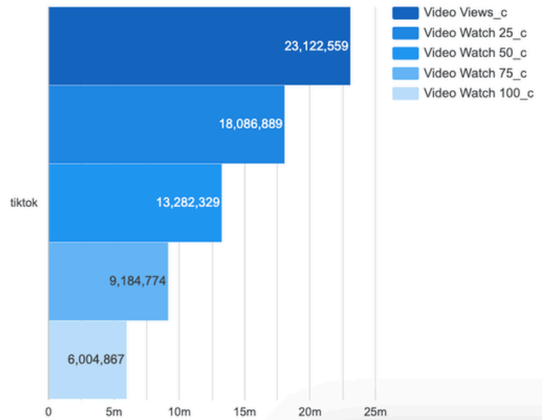



## TikTok

| Video Views | Video Watch 25_c | Video Watch 50_c | Video Watch 75_c | Video Watch 100_c | Likes_c |
|---|---|---|---|---|---|
| 23,122,559 | 18,086,889 | 13,282,329 | 9,184,774 | 6,004,867 | 1,117,850 |

| Shares_c | Comments_c | CTR | CPC | CVR | CPA |
|---|---|---|---|---|---|
| 312,694 | 77,415 | 0.02 | 0.16 | 0.01 | 11 |

### Social Media Campaign Analysis Vertical bar chart

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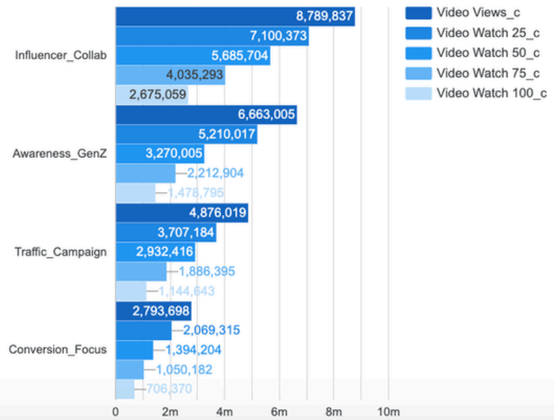

# Conclusion

This project **showcases my ability to transform raw data into a clear, interactive dashboard that supports smarter business decisions**.

I worked on organizing and **analyzing** the dataset, **identifying** key KPIs, and **presenting insights** through effective visualizations.

The result is a **practical reporting solution** that helps stakeholders track performance, uncover trends, and make data-driven decisions with greater confidence.

# Author

## Armando Castillo

*I am a Data Analyst experienced in exploring datasets, performing descriptive and inferential analysis, and translating data into actionable insights for decision-making. My approach combines analytical thinking, attention to detail, and a strong commitment to helping businesses make smarter, data-driven decisions.*